Worksheet 5 · Completing the Square IGCSE

Edexcel IGCSE Mathematics · past paper questions

Name:	Subject: Math	Date:
Learning Objectives: Students will be able to: • Write a quadratic in the form $(x + a)^2 + b$ • Solve quadratic equations by completing the square • Find the turning point of a quadratic graph		
Grade: / 26	**Teacher's Feedback:**	

Answer in the space given. Show your working. Marks for each part are shown in brackets.

Key notes

Formula for x^2+bx+c → $(x+\frac{b}{2})^2-(\frac{b}{2})^2+c$

Formula for ax^2+bx+c → $a(x+\frac{b}{2a})^2-a(\frac{b}{2a})^2+c$

Minimum Point $(x+p)^2+q$ → $(-p,q)$ Symmetry line → $x=-p$

Questions *Completing the Square*

Question 1 (4 marks)

(b) (i) Write $x^2 - 6x + 10$ in the form $(x - a)^2 + b$ where a and b are integers.

(2)

(ii) Hence, describe fully the single transformation that maps the curve with equation $y = x^2$ onto the curve with equation $y = x^2 - 6x + 10$

(2)

Question 2 (3 marks)

(a) Write $3x^2 - 12x + 7$ in the form $a(x + b)^2 + c$

(3)

The line **L** is the line of symmetry of the curve with equation $y = 3x^2 - 12x + 7$

(b) Using your answer to part (a) or otherwise, write down an equation of **L**.

Question 3 (2 marks)

(b) Express $x^2 - 10x + 40$ in the form $(x + a)^2 + b$, where a and b are integers.

(2)

Question 4 (6 marks)

(a) Express $7 - 4x - x^2$ in the form $p - (x + q)^2$ where p and q are constants.

(2)

(b) Use your answer to part (a) to solve the equation $7 - 4(y + 3) - (y + 3)^2 = 0$

Give your solutions in the form $e \pm \sqrt{f}$ where e and f are integers.

(3)

The curve **C** has equation $y = 3 - 5(x + 1)^2$
The point A is the maximum point on **C**.

(c) Write down the coordinates of A.

(.....,.....)

(1)

Question 5 (4 marks)

(a) Express $7 + 12x - 3x^2$ in the form $a + b(x + c)^2$ where a, b and c are integers.

(3)

C is the curve with equation $y = 7 + 12x - 3x^2$
The point A is the maximum point on **C**

(b) Use your answer to part (a) to write down the coordinates of A

(.....,.....)

(1)

(Total for Question 20 is 4 marks)

Question 6 **(4 marks)**

Express each of a, b and c in terms of q so that

$$q + 12x - qx^2$$

can be written as $a - b(x - c)^2$

$a =$ .....

$b =$ .....

$c =$ .....

Question 7 **(3 marks)**

Express each of a, b and c in terms of q so that

$$q + 12x - qx^2$$

can be written as $a - b(x - c)^2$